

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop genuine solutions for the real-world, impactful problems.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: [Márton Vaitkus](#)
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 (**rank 3/34**)
Supervisor: [Galtbayar Artbazar](#)
Thesis: [Evaluating land degradation using image processing methods](#)

EXPERIENCE

Center of Mathematics Application Research Lab Oct, 2023 – June, 2025
Undergraduate Lab Lead & Research Intern

- **Leadership:** Selected as Lab Lead for an 8-student research team; directed weekly seminars to break down complex machine learning and statistical modeling concepts.
- **Data Infrastructure & Pipelines:** Architected scalable data pipelines using **Pandas** and **Dask** to process large-scale sensor datasets; implemented automated workflows with version control and CI/CD integration.
- **Data Infrastructure:** Architected a unified, well-documented data backup infrastructure, streamlining data retrieval and establishing robust protocols for drone imagery processing pipelines.

PROJECTS

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Implemented polynomial regression models from Bishop & Bishop's *Deep Learning Foundations and Concepts* to study **overfitting, underfitting, and the bias-variance tradeoff**.
- Generated synthetic datasets and fit models of varying polynomial degree to analyze how model complexity affects tests and training error.

[aiSim Data Integration for 3D Gaussian Splatting](#)

- Implemented a custom JSON parser to extract and sensor data obtained by aiSim, an autonomous driving simulator, into a nerfstudio (3D representation) format.

AWARDS

Stipendium Hungaricum Scholarship Sep, 2025 – Present

SKILLS

Languages: English (C1 [IELTS 8.0](#)), Mongolian (native), Hungarian (A2)
Programming: Python, Bash, JavaScript, Julia, C/C++, R, MATLAB
Tools: Git, GitHub, Linux, Docker, Pandas, Dask, PyTorch, $\text{\LaTeX}$